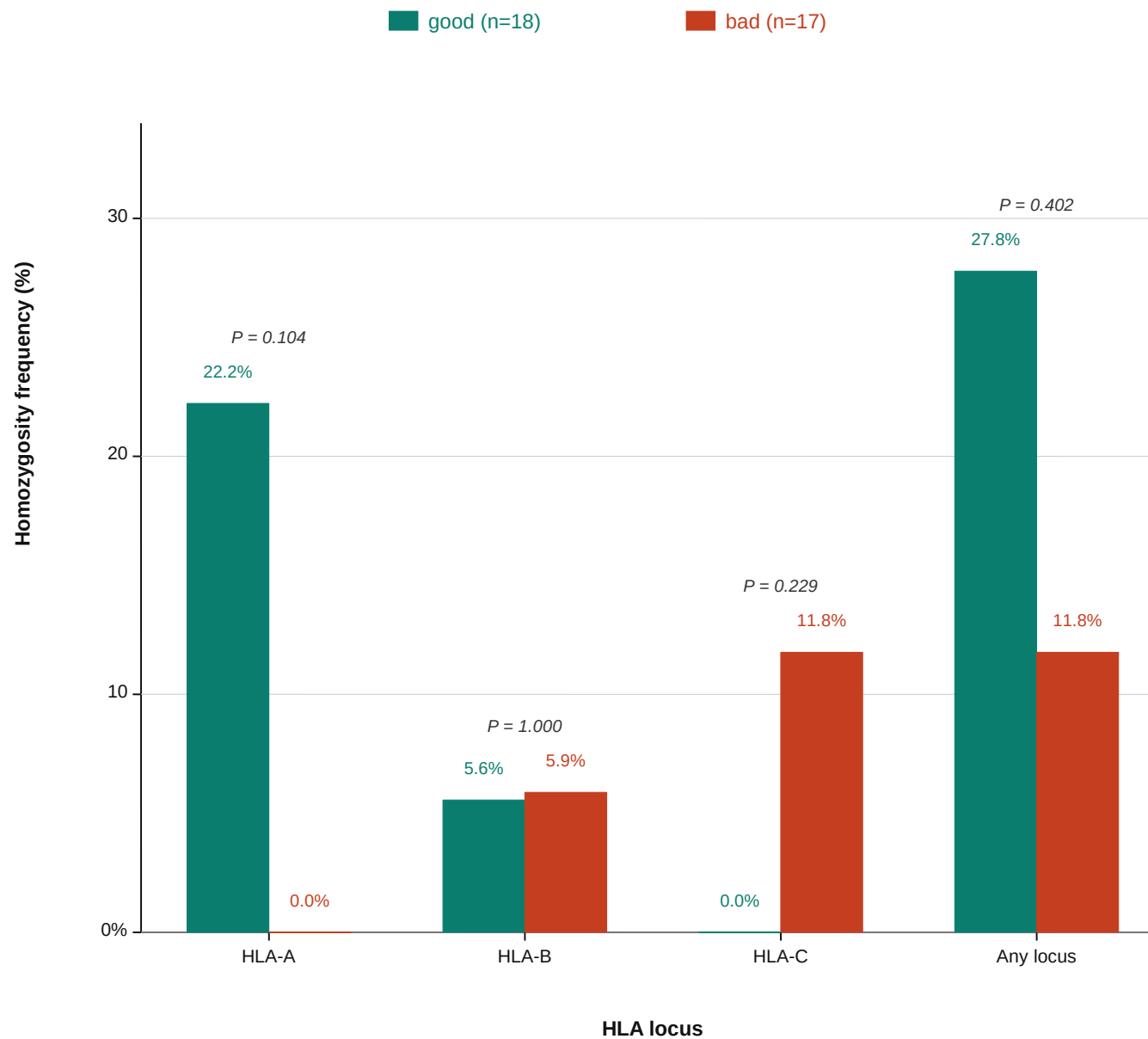


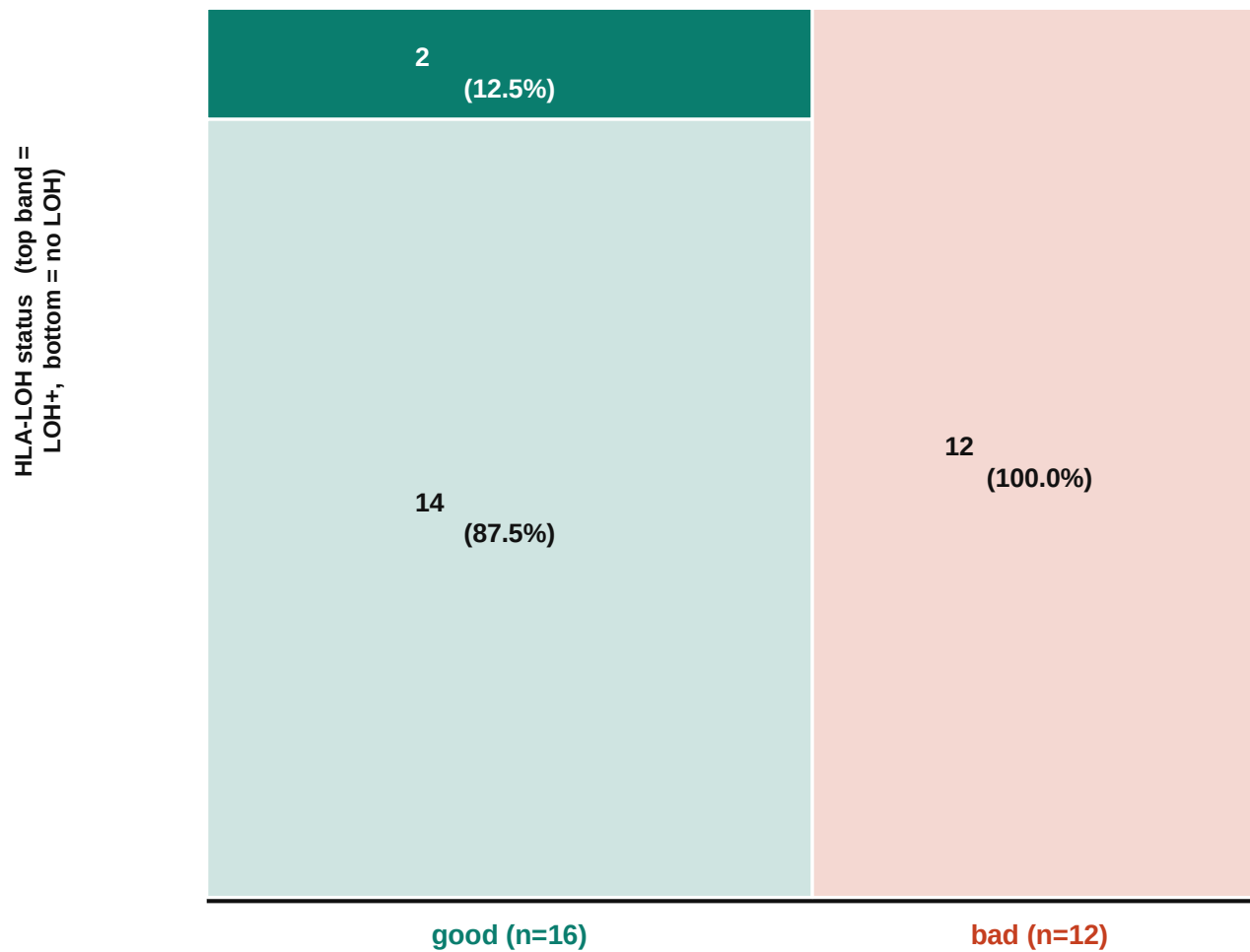
A

Homozygosity per-locus Fisher exact (two-sided). Homozygous HLA reduces neopeptide repertoire diversity.

B

P = 0.492

Fisher exact, two-sided — OR = ∞



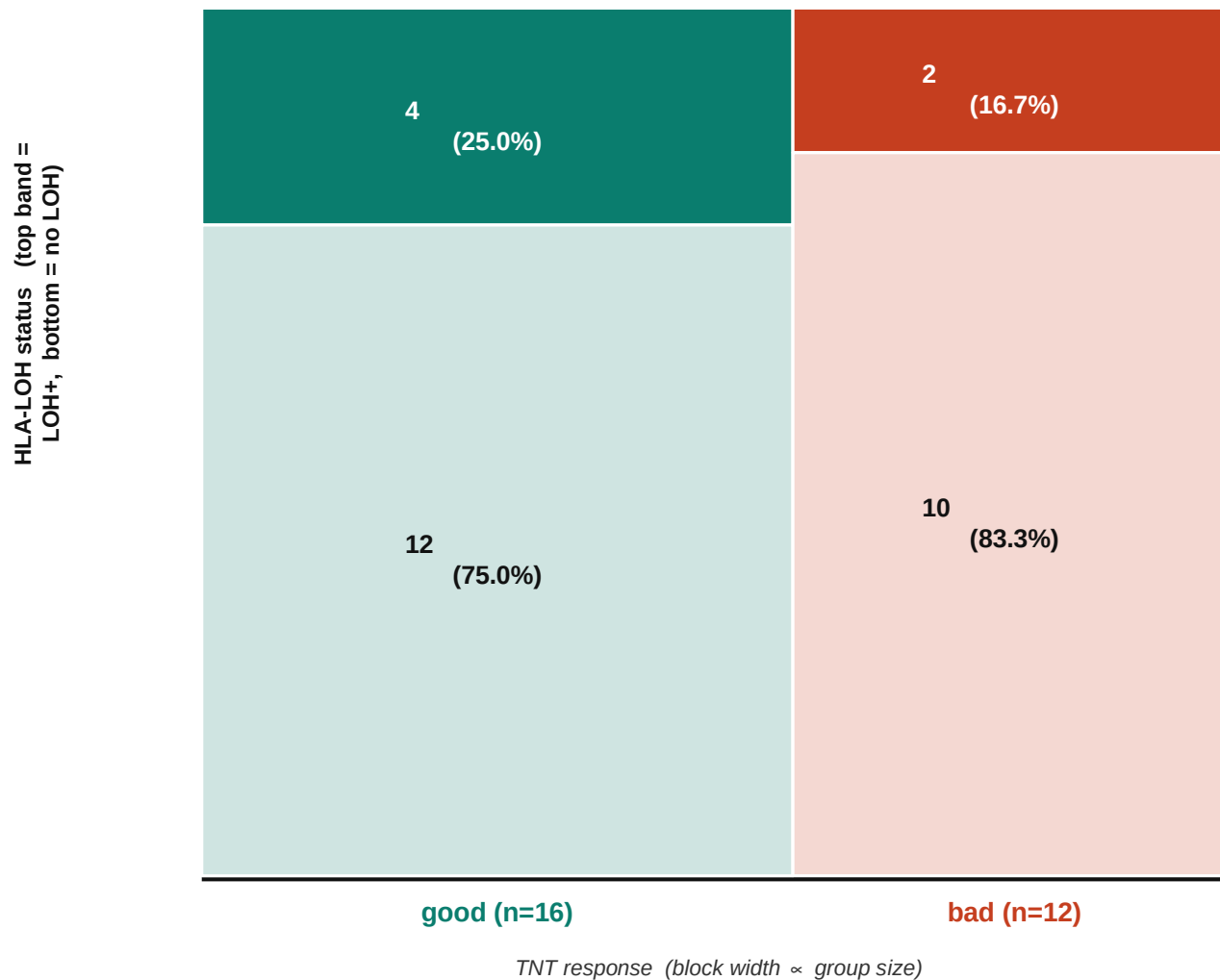
TNT response (block width $\propto$ group size)

Per-subject strict HLA-LOH prevalence (pre-CRT). Good: 2/16, Bad: 0/12.

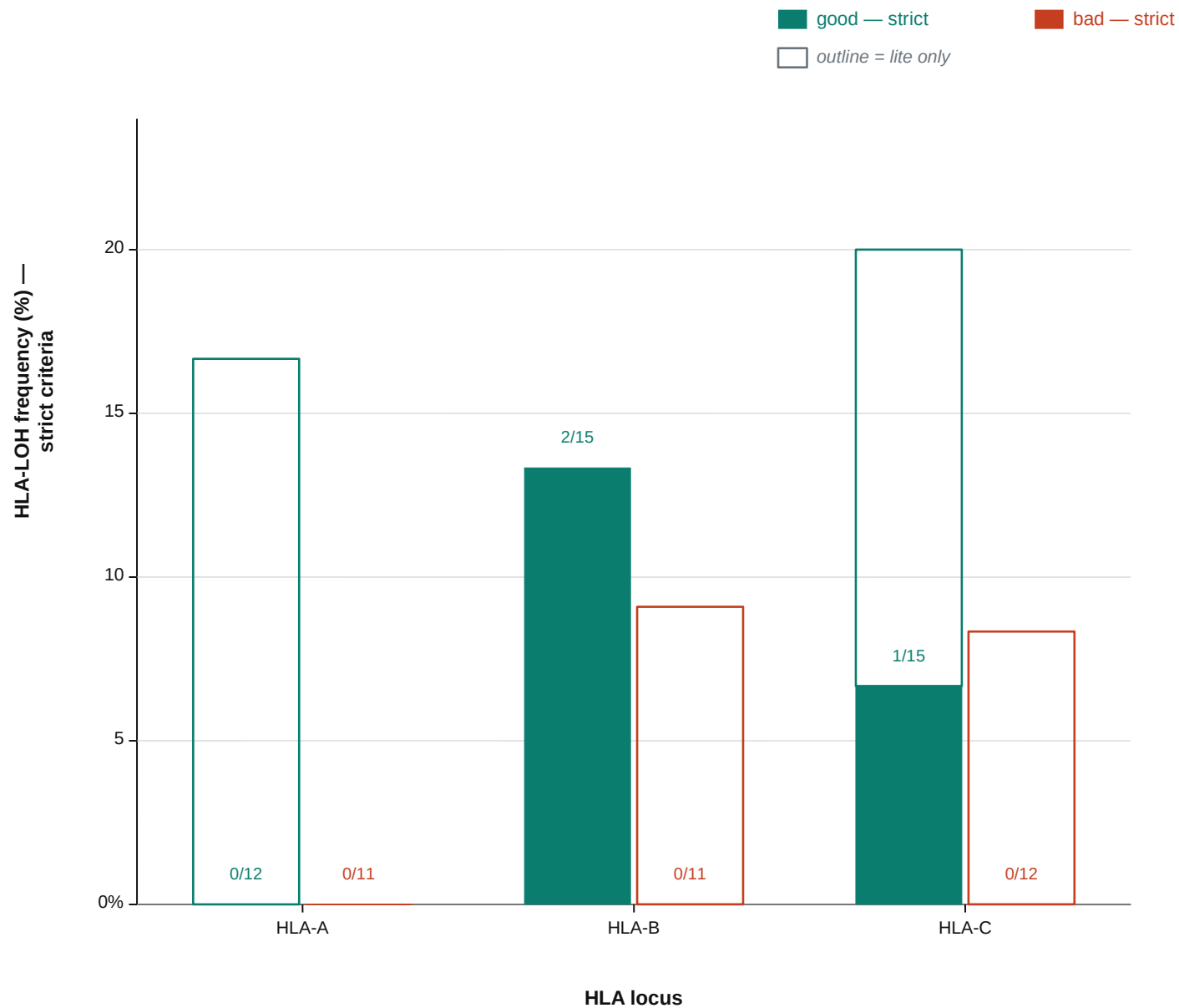
C

P = 0.673

Fisher exact, two-sided — OR = 1.67

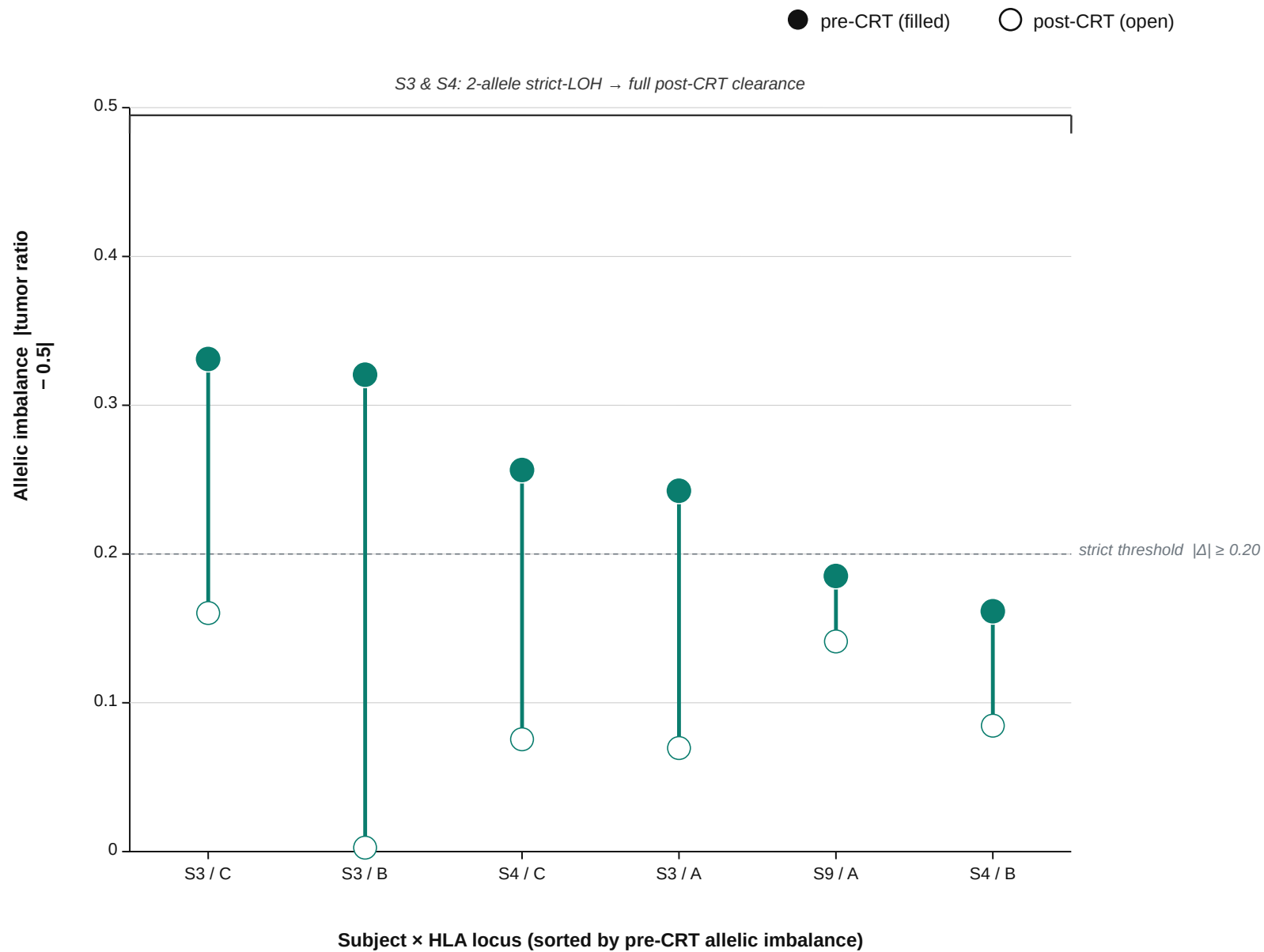


Per-subject LOHHLA-lite HLA-LOH ($|\Delta ratio| \geq 0.15$ + raw Fisher < 0.05). Good: 4/16, Bad: 2/12.

D

Strict: $|\Delta\text{ratio}| \geq 0.20$, Bonferroni-corrected Fisher $P < 0.01$, depth ≥ 30 . Counts annotated as LOH+/het-in-normal.

E



Imbalance = |tumor allele ratio - 0.5|. Filled = pre-CRT, open = post-CRT. N = 6 (subject x locus) pre-CRT LOH-positive events among 13 paired subjects.

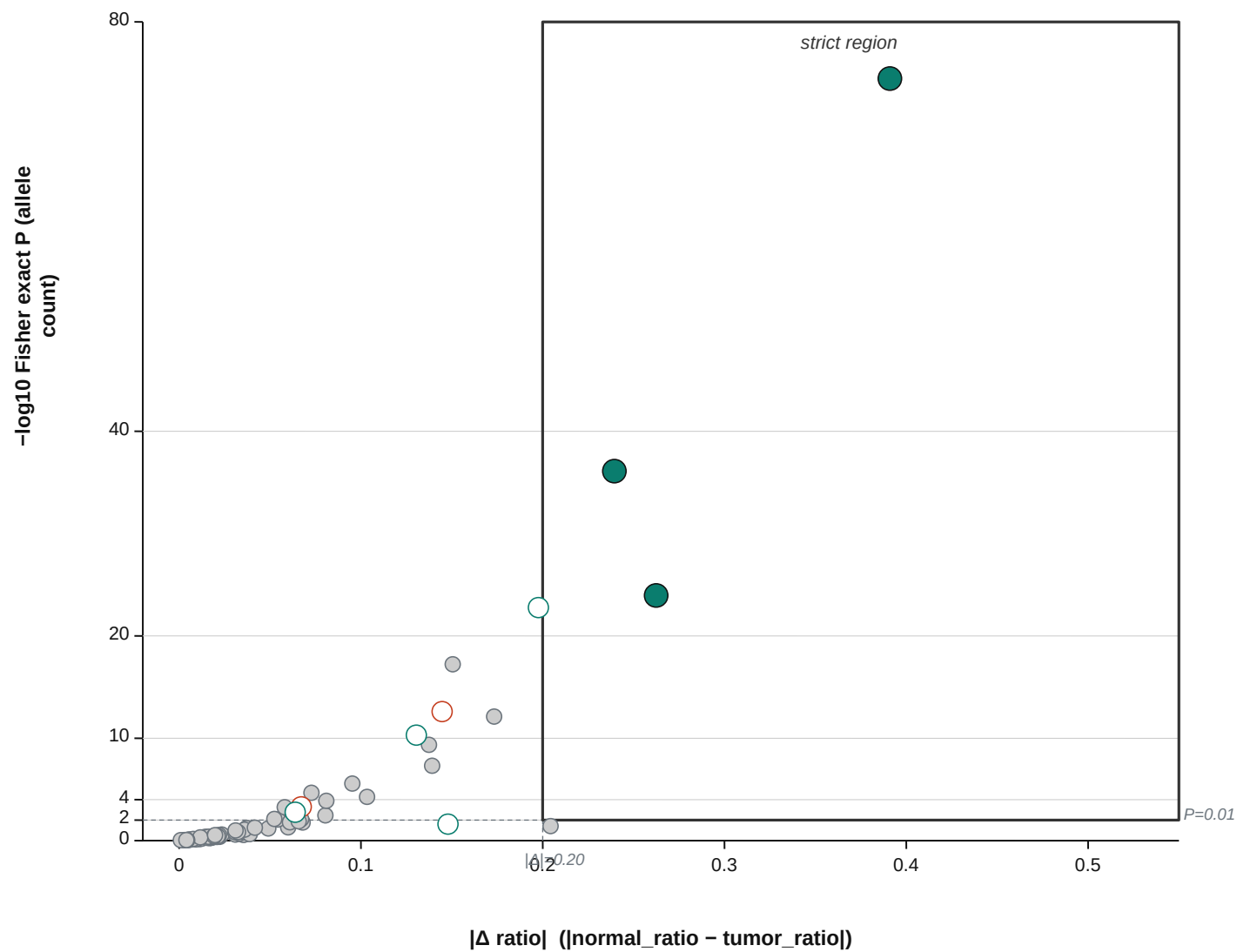
F
● good — strict

● bad — strict

○ good — lite only

○ bad — lite only

● neither (het)



Each point = one (subject, locus) heterozygous event. Strict region defined by both x-threshold and P-threshold.